Notes on consistent model building in (perturbative) QFT

Carlo Marzo

COULD AI DO ALL THIS?

YES

Yes, but

- I already use a hybrid agentic workflow to scan large families of candidate new theories
  (we will try it together at the end)
- It finds interesting results.

Have I published 740 papers in the last week? **No.**

> Most of the work is still human: separating **correct** from incorrect claims, **new** results from redundant ones, **relevant** from irrelevant, **promising** from slop.

Horizon is bigger, human effort unchanged.

Gauge Symmetry Bootstrapping: Building Interactions from Consistency

Gauge bootstrapping, a generalisation of the Noether procedure, starts from a consistent (unitary and causal) free field theory and builds its non-linear, interacting completion order by order, guided only by the requirement that the gauge symmetry be preserved. Remarkably, this minimal procedure recovers the two theories we use to describe physics at the smallest and the largest scales: Yang–Mills theory and Einstein's general relativity.

As a first target, the student will reproduce these classic results at cubic and quartic order:

- non-abelian Yang–Mills theory from n copies of the free Maxwell field;
- the self-interactions of general relativity from the free massless spin-2 field.

. . .

Higher Spins from Reduced Gauge Symmetry: Frame-Like Fields and Fermions

Massless particles of arbitrary spin s are usually described by a symmetric rank-s tensor, together with the requirement of invariance under a gauge symmetry generated by a rank-$(s-1)$ tensor parameter. This framework naturally generalises the U(1) symmetry of electromagnetism and the diffeomorphisms of general relativity. Recent work by our group showed that Fronsdal's action is in fact fixed uniquely by a much weaker requirement: invariance under higher-derivative gauge transformations whose parameter is only a vector.

So far, this result is known only for bosons in Fronsdal's metric-like formulation. The project will test how far it extends. The student will first rederive the result for spin 3 and 4. . .

Interactions with Reduced Higher-Spin Symmetry

Interacting massless particles of spin higher than two are strongly constrained by no-go theorems, from Weinberg's low-energy theorem to the Aragone–Deser obstruction to minimal coupling with gravity. All of these results assume the standard higher-spin gauge symmetry, whose parameter is a rank-$(s-1)$ tensor, as well as a flat background and a finite number of high-spin states. In recent work we have shown that Fronsdal's free action is fixed uniquely by a much weaker symmetry: higher-derivative transformations driven only by a vector parameter. This raises a natural question: if only the reduced symmetry is imposed, how does the interacting sector change?

The student will address this question using the Noether procedure. . .

Understanding our goals is as important as ever. Hence these little notes!

Motivations, definitions, tools

The world as quantum states

The quantum world is a space of possible states: any state is a superposition

$$|\psi\rangle = \sum_n c_n |n\rangle , \qquad c_n = \langle n|\psi\rangle .$$

Time evolution is generated by the Hamiltonian, $i\,\partial_t |\psi(t)\rangle = H |\psi(t)\rangle$.

A clever choice of basis: the energy eigenstates, together with momentum, which commutes with H,

$$H |n\rangle = E_n |n\rangle , \qquad \vec{P} |n\rangle = \vec{p}_n |n\rangle .$$

In this basis the evolution is diagonal:

$$|\psi(t)\rangle = e^{-iHt} |\psi(0)\rangle = \sum_n e^{-iE_n t} c_n |n\rangle .$$

> Diagonalize H and $\vec{P}$, and the dynamics is solved.

Completeness and unitarity

Working assumption: the eigenstates of H span the whole space of states,

$$\sum_n |n\rangle \langle n| = \mathbb{1}.$$

Then the evolution $U(t) = e^{-iHt}$ is unitary, and probability is conserved:

$$\langle\psi(t)|\psi(t)\rangle = \sum_n \left|e^{-iE_n t} \langle n|\psi(0)\rangle\right|^2 = \sum_n \left|\langle n|\psi(0)\rangle\right|^2 = \langle\psi(0)|\psi(0)\rangle\,.$$

> Completeness of the state space and unitarity go together: no state can be missing from the sum.

Relativistic invariance: the Poincaré group

Translations P^μ and Lorentz transformations $J^{\mu\nu}$ act on the states, with

$$
\begin{aligned}
[P^\mu, P^\nu] &= 0, \\
[P^\mu, J^{\rho\sigma}] &= i\left(\eta^{\mu\rho} P^\sigma - \eta^{\mu\sigma} P^\rho\right), \\
[J^{\mu\nu}, J^{\rho\sigma}] &= i\left(\eta^{\nu\rho} J^{\mu\sigma} - \eta^{\mu\rho} J^{\nu\sigma} - \eta^{\nu\sigma} J^{\mu\rho} + \eta^{\mu\sigma} J^{\nu\rho}\right).
\end{aligned}
$$

Energy is not invariant, but two operators commute with all generators (Casimir invariants):

$$
P^2 = P_\mu P^\mu = m^2, \qquad W^2 = W_\mu W^\mu = -m^2\, s(s+1), \qquad W_\mu = \tfrac{1}{2}\,\epsilon_{\mu\nu\rho\sigma} J^{\nu\rho} P^\sigma,
$$

where W_μ is the Pauli–Lubanski vector; for $m = 0$, $W^\mu = h\, P^\mu$ with helicity h.

> The invariant job: classify all possible states by the Casimirs, mass $P^2 = m^2$ instead of energy, and spin s (helicity for massless states).

The solvable case: free particles

The simplest states: one free particle of mass m and spin s, with $\sigma = -s, \ldots, s$,

$$\vec{P}\,|\vec{p},\sigma\rangle = \vec{p}\;|\vec{p},\sigma\rangle\,, \qquad H\,|\vec{p},\sigma\rangle = E_p\,|\vec{p},\sigma\rangle\,, \qquad E_p = \sqrt{\vec{p}^{\,2}+m^2}.$$

Completeness, with the relativistic normalization $\langle\vec{p},\sigma|\vec{p}\,',\sigma'\rangle = 2E_p\,(2\pi)^3\,\delta^3(\vec{p}-\vec{p}\,')\,\delta_{\sigma\sigma'}$:

$$\sum_\sigma \int \frac{d^3p}{(2\pi)^3\,2E_p}\;|\vec{p},\sigma\rangle\,\langle\vec{p},\sigma| = \mathbb{1}_{1\text{ particle}}.$$

Time evolution is a phase for each momentum:

$$|\psi(t)\rangle = \sum_\sigma \int \frac{d^3p}{(2\pi)^3\,2E_p}\;e^{-iE_pt}\,\psi_\sigma(\vec{p})\;|\vec{p},\sigma\rangle\,, \qquad \psi_\sigma(\vec{p}) = \langle\vec{p},\sigma|\psi(0)\rangle\,.$$

Many free particles: $H\,|\vec{p}_1\sigma_1,\ldots,\vec{p}_k\sigma_k\rangle = (E_{p_1}+\cdots+E_{p_k})\,|\vec{p}_1\sigma_1,\ldots,\vec{p}_k\sigma_k\rangle$.

> The free theory is diagonal from the start: no fields needed, only particle states.

From states to fields

- This approach is explicitly unitary: a complete set of states, evolving with e^{-iHt}.
- As we discussed, this is not how we work. We introduce field-valued Hamiltonians, or Lagrangians $\mathcal{L}(\phi, \partial\phi)$, to enforce locality and causality:

$$[\phi(x), \phi(y)] = 0 \qquad \text{for } (x-y)^2 < 0.$$

How do we “solve” the evolution in the presence of local fields?

The free theory in momentum space

Fourier transform the field; derivatives become momenta:

$$\phi(x) = \int \frac{d^4p}{(2\pi)^4}\, e^{-ip\cdot x}\, \tilde{\phi}(p), \qquad \partial_\mu \longrightarrow -ip_\mu.$$

The action of the free scalar becomes a double integral over momenta:

$$S = \int d^4x \left(\tfrac{1}{2}\, \partial_\mu\phi\, \partial^\mu\phi - \tfrac{1}{2}\, m^2\phi^2 \right) = \frac{1}{2} \int \frac{d^4p\, d^4p'}{(2\pi)^8} \left(-p\cdot p' - m^2 \right) \tilde{\phi}(p)\, \tilde{\phi}(p') \underbrace{\int d^4x\, e^{-i(p+p')\cdot x}}_{(2\pi)^4\, \delta^4(p+p')}.$$

Integrating the Lagrangian over spacetime imposes momentum conservation, $p' = -p$:

$$S = \frac{1}{2} \int \frac{d^4p}{(2\pi)^4}\, \tilde{\phi}(-p)\, (p^2 - m^2)\, \tilde{\phi}(p).$$

> Each momentum mode decouples: the quadratic operator becomes a number, $p^2 - m^2$, for each p.

Solving the free theory: the propagator

- In terms of fields, the particle states are hidden in the background: the field creates and destroys them.
- The free theory is a quadratic Lagrangian; in momentum space its operator is $p^2 - m^2$.
- "Solving" it means inverting the quadratic operator: the propagator

$$D(p) = \frac{i}{p^2 - m^2 + i\epsilon}.$$

The free theory is solved by computing the propagator. The particle states reappear as its pole, at $p^2 = m^2$.

Propagators and the space of states

- We traded the diagonal, trivial time evolution of particle states for the propagator.
  For the free scalar, $D(p) = \dfrac{i}{p^2 - m^2 + i\epsilon}$.
- In the full, interacting theory the states are still there, classified by the Poincaré Casimirs:

$$|\Omega\rangle \text{ (vacuum)}, \qquad |\vec{p}\rangle \text{ (one particle)}, \qquad |\vec{p}_1, \vec{p}_2, \ldots\rangle \text{ (many particles)}.$$

How are the properties of the propagator connected with the properties of the space of states?

Poles create states

Back to the free scalar. In momentum space the equation of motion is

$$\left(p^2 - m^2\right) \tilde{\phi}(p) = 0,$$

which has non-zero solutions only on the mass shell, $p^2 = m^2$: those are the particle states.

The propagator is the inverse of the same operator,

$$D(p) = \frac{i}{p^2 - m^2 + i\epsilon},$$

so its pole sits exactly where the equation of motion has solutions.

> Zeros of the equation of motion in momentum space $=$ poles of the propagator $=$ states with a given p^2.

A generic vector field

The most general quadratic Lagrangian of a vector, with no gauge symmetry (everything propagates):

$$\mathcal{L} = k_1\,\partial_\mu V_\nu\,\partial^\mu V^\nu + k_2\,(\partial_\mu V^\mu)^2 + m_1\,V_\mu V^\mu.$$

In momentum space, split V into its transverse (spin 1) and longitudinal (spin 0) parts,

$$\theta_{\mu\nu} = \eta_{\mu\nu} - \frac{p_\mu p_\nu}{p^2}, \qquad \omega_{\mu\nu} = \frac{p_\mu p_\nu}{p^2}, \qquad \theta + \omega = \mathbb{1}:$$

$$\mathcal{L} = \tfrac{1}{2}\,\tilde{V}^\mu(-p)\Big[\underbrace{2\big(k_1 p^2 + m_1\big)}_{\text{spin 1}}\,\theta_{\mu\nu} + \underbrace{2\big(\big(k_1+k_2\big)\,p^2 + m_1\big)}_{\text{spin 0}}\,\omega_{\mu\nu}\Big]\tilde{V}^\nu(p).$$

Why projectors: inversion becomes trivial

θ and ω are projectors: they are orthogonal and add up to the identity,

$$\theta^2 = \theta, \qquad \omega^2 = \omega, \qquad \theta\,\omega = \omega\,\theta = 0, \qquad \theta + \omega = \mathbb{1}.$$

So an operator written as $K = A\,\theta + B\,\omega$ is inverted coefficient by coefficient:

$$K^{-1} = \frac{1}{A}\,\theta + \frac{1}{B}\,\omega, \qquad \text{since} \qquad K\,K^{-1} = \theta + \omega = \mathbb{1}.$$

$$\text{generic vector:} \qquad D_{\mu\nu}(p) = i\left[\frac{\theta_{\mu\nu}}{2\left(k_1 p^2 + m_1\right)} + \frac{\omega_{\mu\nu}}{2\left((k_1 + k_2)\,p^2 + m_1\right)}\right],$$

with poles at $p^2 = -m_1/k_1$ (spin 1) and $p^2 = -m_1/(k_1 + k_2)$ (spin 0).

> Projectors turn inverting a tensor operator into dividing by numbers. That is why they are sought after, and why they are at the core of automatic spectrographs such as `PSALTer` and `Kummitus`.

The residues

Near each pole the propagator behaves as residue$/(p^2 - M^2)$. For the two sectors:

	states	**mass squared** M^2	**residue**
spin 1	3	$-\frac{m_1}{k_1}$	$-\frac{1}{2k_1}$
spin 0	1	$-\frac{m_1}{k_1+k_2}$	$\frac{1}{2(k_1+k_2)}$

We cannot accept tachyons ($M^2 < 0$), which propagate faster than light. Both masses squared positive requires k_1 and $k_1 + k_2$ to have the same sign, and then the two residues have opposite signs.

The only way out: $k_2 = -k_1$, so that the spin-0 operator has no p^2 and no pole. That is Proca, $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}m^2 V_\mu V^\mu$.

> No tachyons $\Longrightarrow$ residues of opposite sign. Why this is a problem: later.

The same computation with `Kummitus`

In

```
ConstantSymbols /@ {k1, k2, m1};
model = k1 dd[-a][V[-b]] dd[a][V[b]] + k2 dd[-a][V[a]] dd[-b][V[b]] + m1 V[-a] V[a];
SourceDeclare[model];  CompSatProp[model];
masses
ComputeMassiveStates[]
```

Out

sectors 1^- and 0^+: $a(1^-) = 2\,(m_1 + k_1\,q^2)$, $a(0^+) = 2\,(m_1 + (k_1 + k_2)\,q^2)$

poles: $\left\{ -\frac{m_1}{k_1},\ -\frac{m_1}{k_1+k_2} \right\}$

residues: $\left\{ \left\{ -\frac{m_1}{k_1},\ \left\{ -\frac{1}{2k_1},\ -\frac{1}{2k_1},\ -\frac{1}{2k_1} \right\} \right\},\ \left\{ -\frac{m_1}{k_1+k_2},\ \left\{ \frac{1}{2(k_1+k_2)} \right\} \right\} \right\}$

The same sectors, poles and residues as by hand: three states at the spin-1 pole, one at the spin-0 pole.

One thing you need to accept

The propagator, the inverse of the quadratic operator, is the vacuum expectation value of the time-ordered product of two fields:

$$\langle 0|T\,\phi(x)\,\phi(y)|0\rangle \;=\; D_F(x-y) \;=\; \int \frac{d^4p}{(2\pi)^4}\,\frac{i\,e^{-ip\cdot(x-y)}}{p^2-m^2+i\epsilon}.$$

It is the amplitude for a state created at y to be found at x: this is how the propagator talks to the states.

Exercise: free scalar, canonical quantization

1. Expand $\phi(x) = \int \frac{d^3p}{(2\pi)^3\,2E_{\vec{p}}}\left(a_{\vec{p}}\,e^{-ip\cdot x} + a^\dagger_{\vec{p}}\,e^{ip\cdot x}\right)$ and compute $\langle 0|\phi(x)\phi(y)|0\rangle$.
2. Write the time ordering with step functions, $\theta(x^0-y^0)$ and $\theta(y^0-x^0)$.
3. Show that the p^0 integral, closed in the complex plane, gives the $+i\epsilon$ prescription.

Why the sign of the residues matters

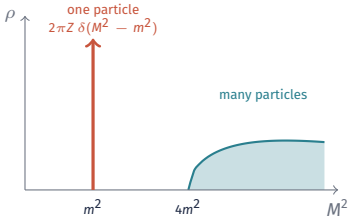
Insert the completeness of the states of the full theory between the fields,

$$\mathbb{1} = |\Omega\rangle\langle\Omega| + \sum_\lambda \int \frac{d^3p}{(2\pi)^3\, 2E_p(\lambda)}\, |\lambda_{\vec{p}}\rangle\langle\lambda_{\vec{p}}|\,,$$

and use translations, then a Lorentz boost to the rest frame:

$$\begin{aligned}\langle\Omega|\phi(x)|\lambda_{\vec{p}}\rangle &= \langle\Omega|\, e^{iP\cdot x}\,\phi(0)\, e^{-iP\cdot x}\,|\lambda_{\vec{p}}\rangle = e^{-ip\cdot x}\,\langle\Omega|\phi(0)|\lambda_{\vec{p}}\rangle \\ &= e^{-ip\cdot x}\,\langle\Omega|\phi(0)|\lambda_{\vec{0}}\rangle\,.\end{aligned}$$

$$\begin{aligned}\langle\Omega|T\phi(x)\phi(y)|\Omega\rangle &= \int_0^\infty \frac{dM^2}{2\pi}\,\rho(M^2)\,D_F(x-y;\,M^2), \\ \rho(M^2) &= \sum_\lambda 2\pi\,\delta(M^2-m_\lambda^2)\,\big|\,\langle\Omega|\phi(0)|\lambda_{\vec{0}}\rangle\,\big|^2.\end{aligned}$$



Cumbersome but straightforward derivation: see the Källén–Lehmann spectral representation in a textbook, e.g. Peskin & Schroeder or Schwartz.

In momentum space: $D(p) = \int_0^\infty \frac{dM^2}{2\pi}\,\rho(M^2)\,\frac{i}{p^2-M^2+i\epsilon} = \frac{iZ}{p^2-m^2+i\epsilon} + \dots$

What the propagator tells us about the states

Completeness and positive norms of the states give a positive spectral function:

$$\rho(M^2) \geq 0, \qquad \text{in particular} \qquad Z = \left| \langle \Omega | \phi(0) | \vec{p} \rangle \right|^2 \geq 0.$$

Propagator	**Space of states**
pole at $p^2 = m^2$	a particle of mass m; $m^2 \geq 0$, no tachyons
residue Z at a single pole	norm of the state: $Z > 0$ physical, $Z < 0$ ghost
number of independent residues	number of propagating states
falls no faster than $1/p^2$ at large p	follows from $\rho \geq 0$

Unitarity is not manifest with fields, but it leaves its fingerprint on the propagator: real poles with positive residues.

Back to Kummitus: removing the spin-0 sector

Choose $k_2 = -k_1$: the spin-0 operator loses its p^2.

In

```
model = k1 (dd[-a][V[-b]] dd[a][V[b]] - dd[-a][V[a]] dd[-b][V[b]]) + m1 V[-a] V[a];
SourceDeclare[model];  CompSatProp[model];
masses
ComputeMassiveStates[]
```

Out

sectors: $a(1^-) = 2\,(m_1 + k_1\,q^2)$, $a(0^+) = 2\,m_1$ (no pole)

poles: $\left\{ -\frac{m_1}{k_1} \right\}$

residues: $\left\{ \left\{ -\frac{m_1}{k_1},\ \left\{ -\frac{1}{2k_1},\ -\frac{1}{2k_1},\ -\frac{1}{2k_1} \right\} \right\} \right\}$

One pole, three states with residues of the same sign. Positive mass squared and positive residues for $k_1 < 0$, $m_1 > 0$: a healthy massive spin 1, Proca.

Back to Kummitus: removing the spin-1 sector

Choose $k_1 = 0$: the spin-1 operator loses its p^2.

In

```
model = k2 dd[-a][V[a]] dd[-b][V[b]] + m1 V[-a] V[a];
SourceDeclare[model];  CompSatProp[model];
masses
ComputeMassiveStates[]
```

Out

sectors: $a(1^-) = 2\,m_1$ (no pole), $a(0^+) = 2\,(m_1 + k_2\,q^2)$

poles: $\left\{ -\frac{m_1}{k_2} \right\}$

residues: $\left\{ \left\{ -\frac{m_1}{k_2},\ \left\{ \frac{1}{2k_2} \right\} \right\} \right\}$

One pole, one state. Positive mass squared and positive residue for $k_2 > 0$, $m_1 < 0$: a healthy massive spin 0, carried by a vector field.

Using `xAct` for model building

Mastering ansätze and functional derivatives

- An ansatz: the most general Lagrangian for a given field content, with one free coefficient for each independent contraction, as k_1, k_2, m_1 for the vector.
- With `xAct`, the computer does the bookkeeping:
  - define the fields, `DefTensor`;
  - generate all the independent contractions, `xTras`;
  - take functional derivatives, `VarD`, to get the equations of motion;
  - go to momentum space and read off the quadratic operator.

→ live in the notebook

Gauge symmetries and interactions

Next time!